

Compare and contrast Mitosis and Meiosis Worksheet

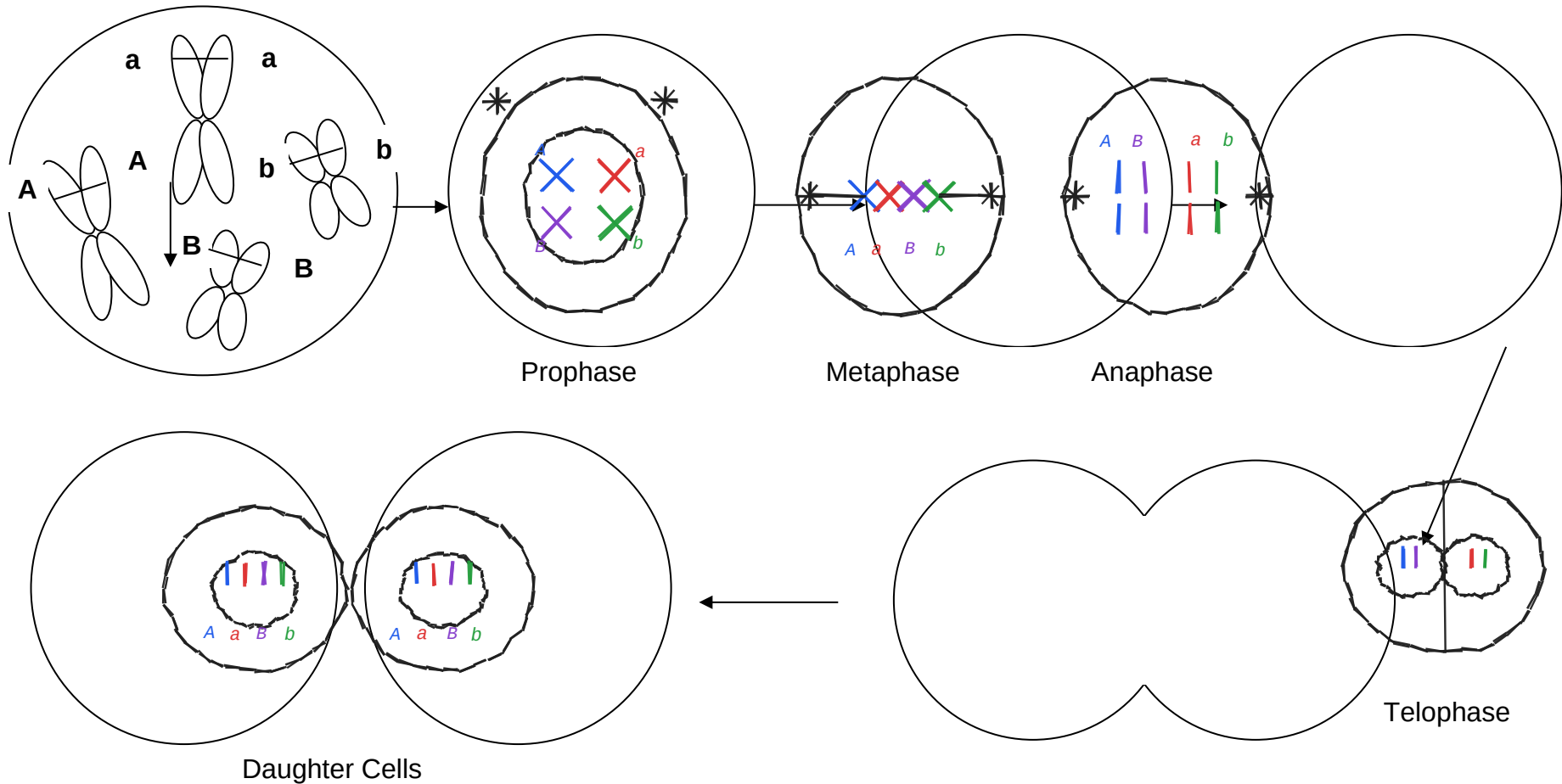
Part I Mitosis→ Use Diploid skin cells for Mitosis with $2n = 4$ (as shown in picture below)

Material:

colored pencils

- blue and purple for dad's chromosomes (genes A, B)
- red and green from mom's chromosomes (genes a, b)

- I. Draw the chromosomes in the cell as it undergoes **Mitosis**:
Remember homologous chromosomes line up on separate spindle fiber



Part II Meiosis

II. Draw the chromosomes in the cell as it undergoes Meiosis (same parent cell):

Use Use Diploid testical cells for Mitosis with $2n = 4$ (as shown in the picture below)

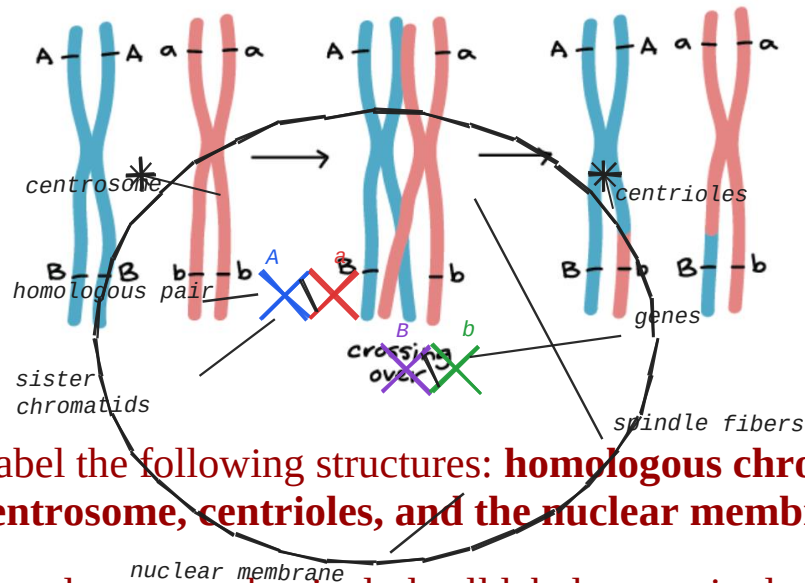
Remember Homologous chromosomes line up on separate spindle fibers

Material:

colored pencils

- blue and purple for dad's chromosomes (genes A, B)
- red and green from mom's chromosomes (genes a, b)

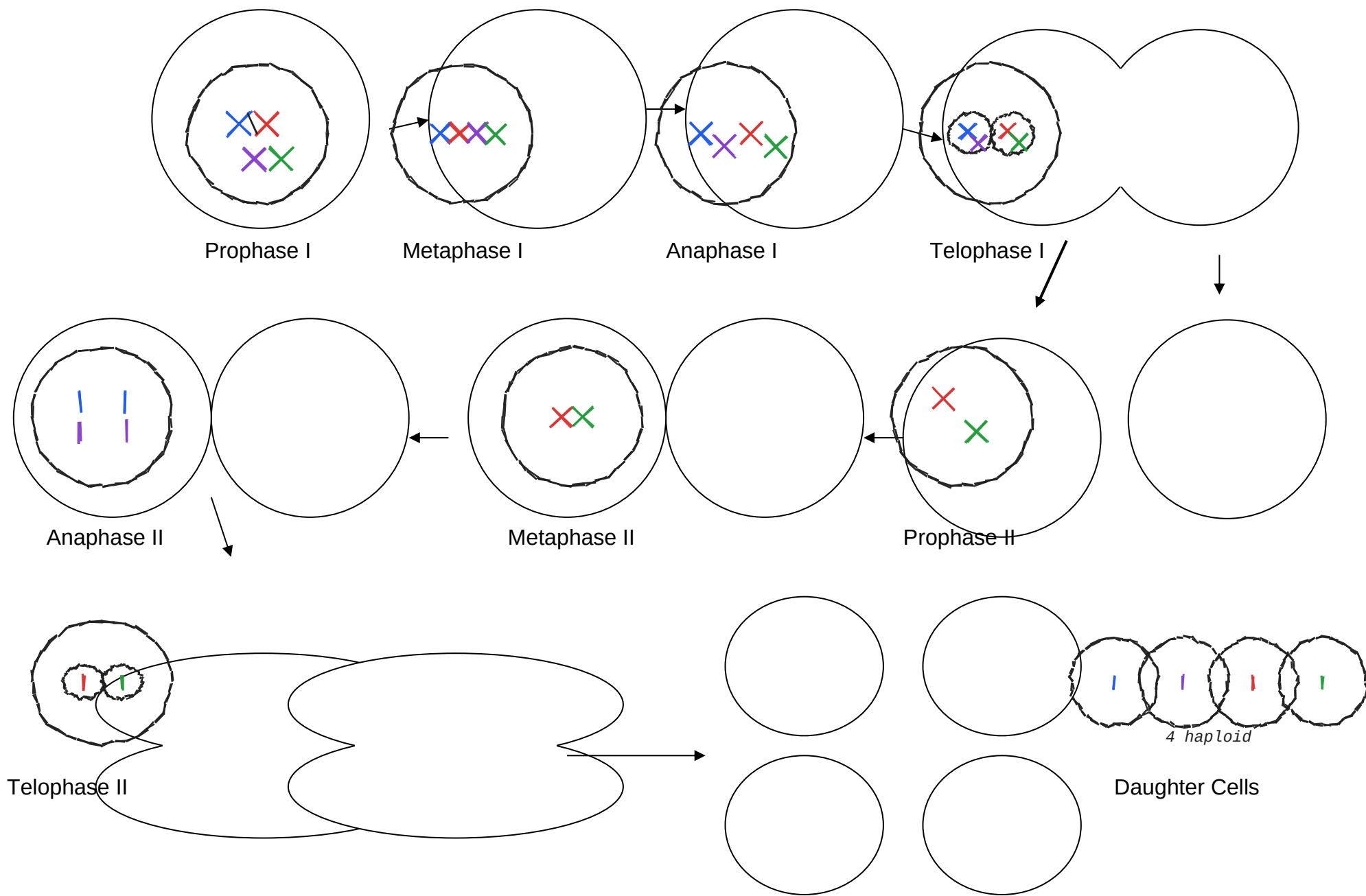
Show crossing over by switching the genes during Prophase I of meiosis



Note that chromosomes A, a are on one homologous pair of chromosomes in our exercise, and B, b are on a separate second homologous pair of chromosomes

Label the following structures: **homologous chromosomes, sister chromatids, genes, spindle fibers, centrosome, centrioles, and the nuclear membrane.**

You do **not** need to include all labels on a single cell—spread them across the diagram so each structure is clearly visible and easy to identify.



Part III: Complete the summary chart Comparing and Contrasting Mitosis and Meiosis

Cell Processes	Mitosis	Meiosis
Creates	<i>body cells</i>	<i>sex cells (gametes)</i>
Definition	<i>1 division -> 2 identical 2n cells</i>	<i>2 divisions -> 4 different n cells</i>
End Products	<i>2 diploid cells</i>	<i>4 haploid cells</i>
Steps	<i>PMAT once</i>	<i>PMAT I + PMAT II</i>
Type of Reproduction	<i>asexual</i>	<i>sexual</i>

Are they identical to the parent cell?	<i>yes</i>	<i>no</i>
When does cytokinesis occur?	<i>once (after telophase)</i>	<i>twice (after I and II)</i>
How many times does the parent cell divide?	<i>1 time</i>	<i>2 times</i>
What happens to the number of chromosomes at the end of each process? Are they in pairs or individual chromosomes?	<i>stays $2n$ (46 in humans) in pairs</i>	<i>halves to n (23 in humans) individual</i>
Why is each important?	<i>growth + repair</i>	<i>variation + proper # for fertilization</i>

How many chromosomes do human body cells and human sex cells have after they go through each process?	<i>Mitosis: body cells = 46</i> <i>Meiosis: sex cells = 23</i>	
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Part III

Explain **nondisjunction**—when chromosomes or chromatids fail to separate properly—and how it can lead to conditions such as **Down syndrome** and **Turner syndrome**.

Then explain how these chromosomal abnormalities result in different physical and developmental effects.

Nondisjunction = chromosomes do not separate right.
In meiosis I homologous chromosomes can fail to split;
meiosis II sister chromatids can fail to split.
This makes gametes with extra or missing chromosomes.
Down syndrome: trisomy 21 (47 total).
Turner syndrome: monosomy X (45, X).
These change gene dosage, so development can differ:
Down can include learning delays and heart risks;
Turner can include short stature and ovarian issues.